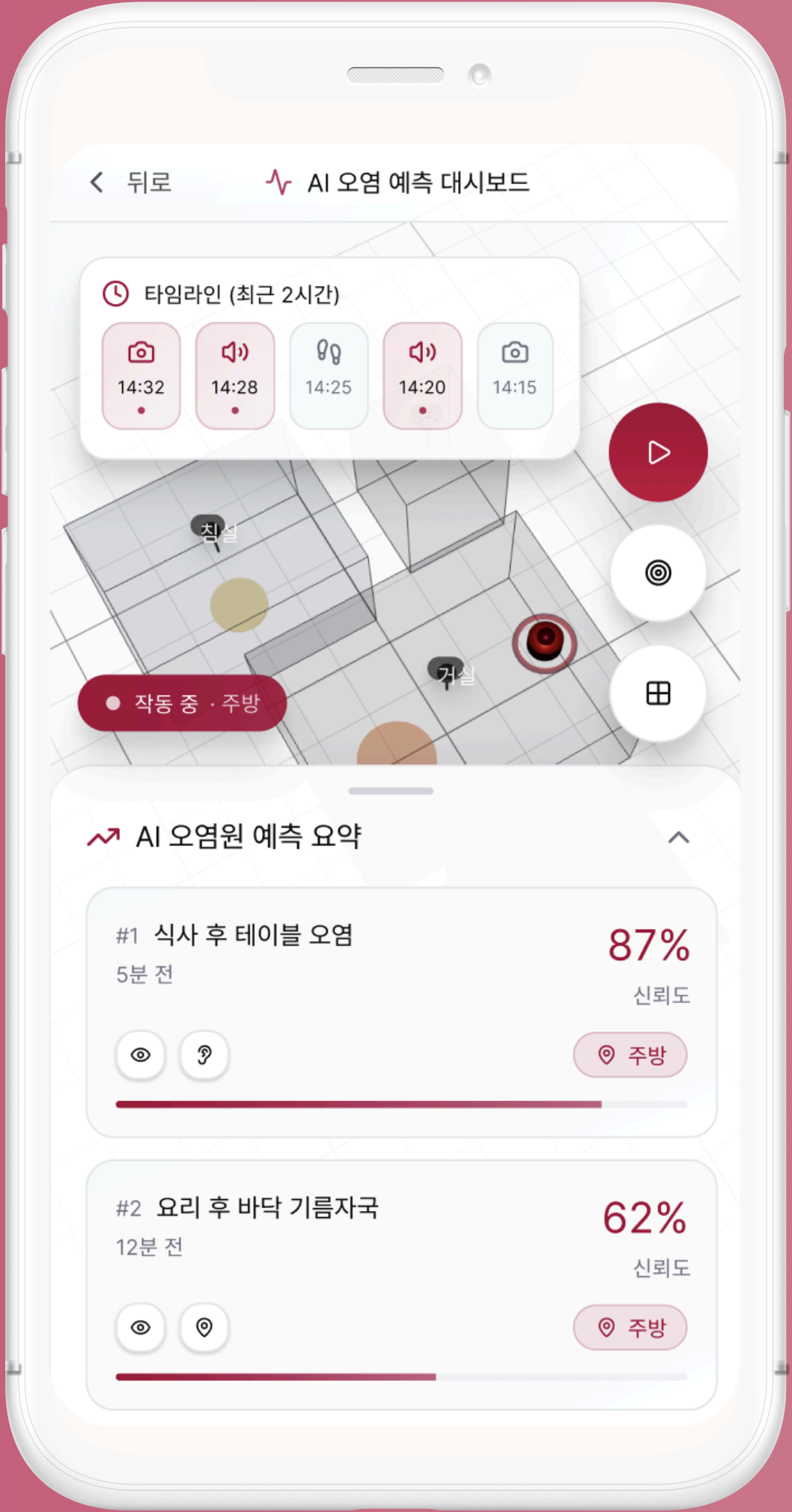
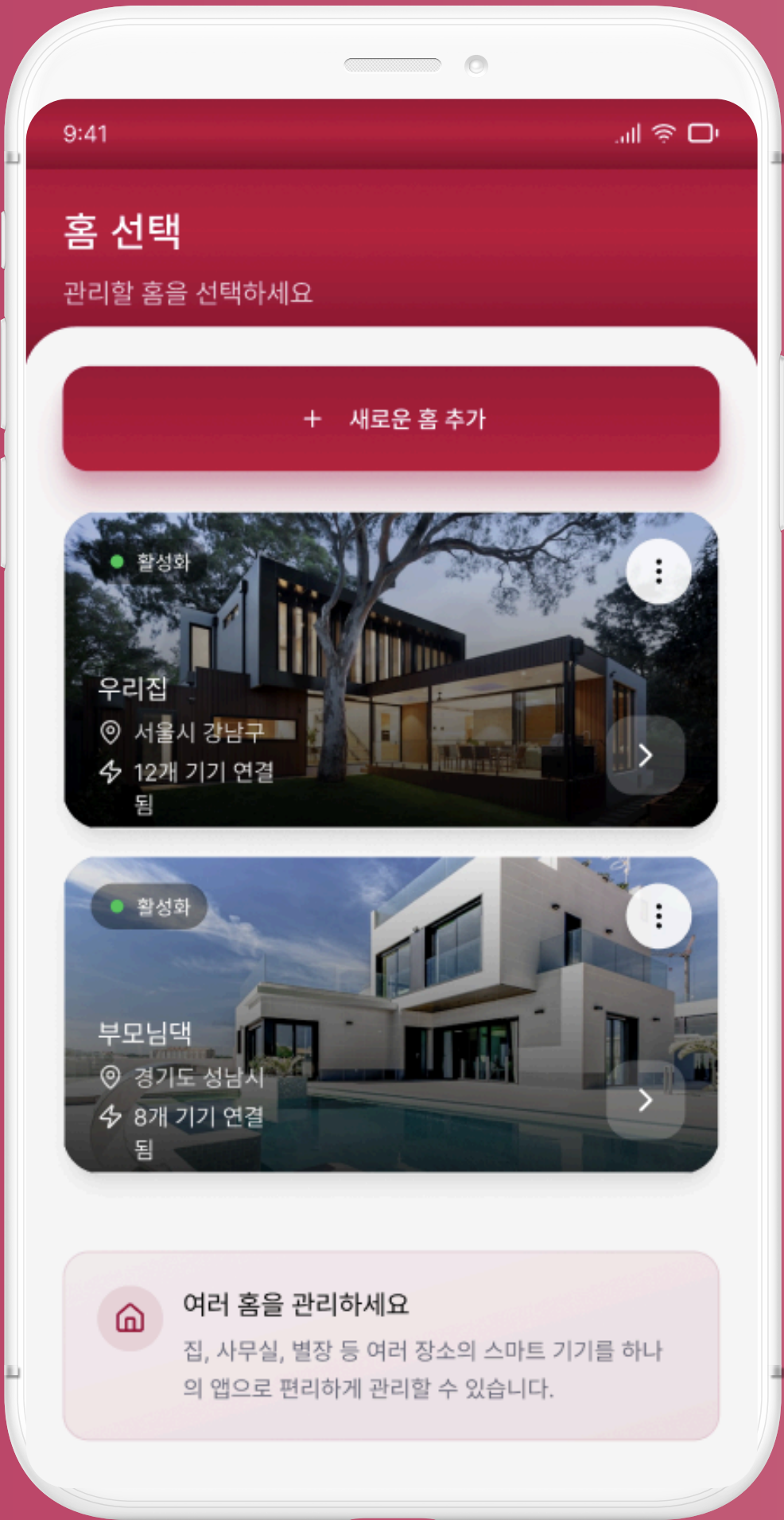


LOCUS

사용자의 환경을 인지해 스스로 오염원을
파악하여 청소하는 지능형 로봇청소기 시스템



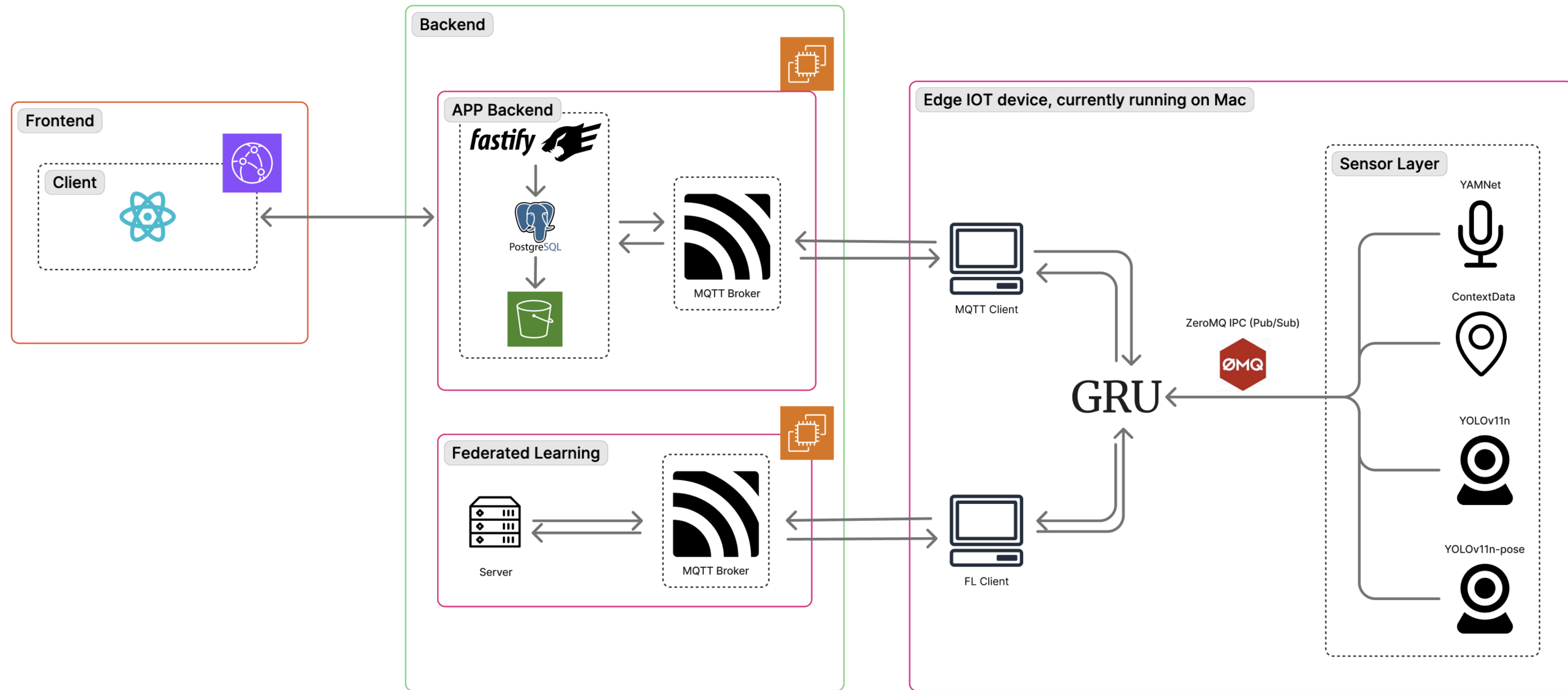
Introduction

Problem : 로봇청소기는 예정된 청소, 혹은 명령받은 청소밖에 하지 못한다

Question : 로봇청소기에게 지능을 부여한다면?

IDEA : 로봇청소기에게 멀티모달(시각, 청각, 공간, 시간) 데이터를 학습시켜,
직접 언제 어디를 청소할지 판단하게 합니다.

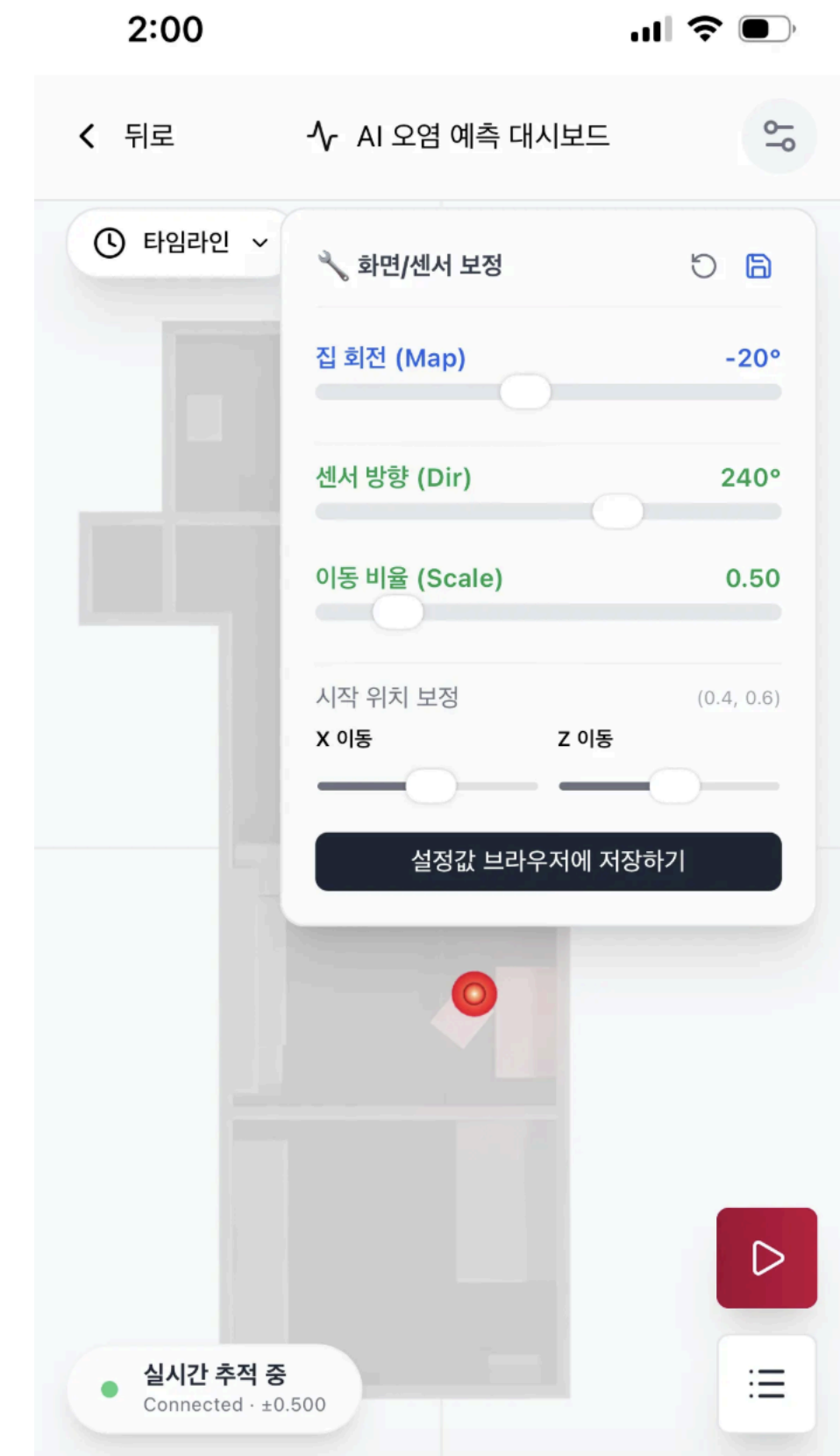
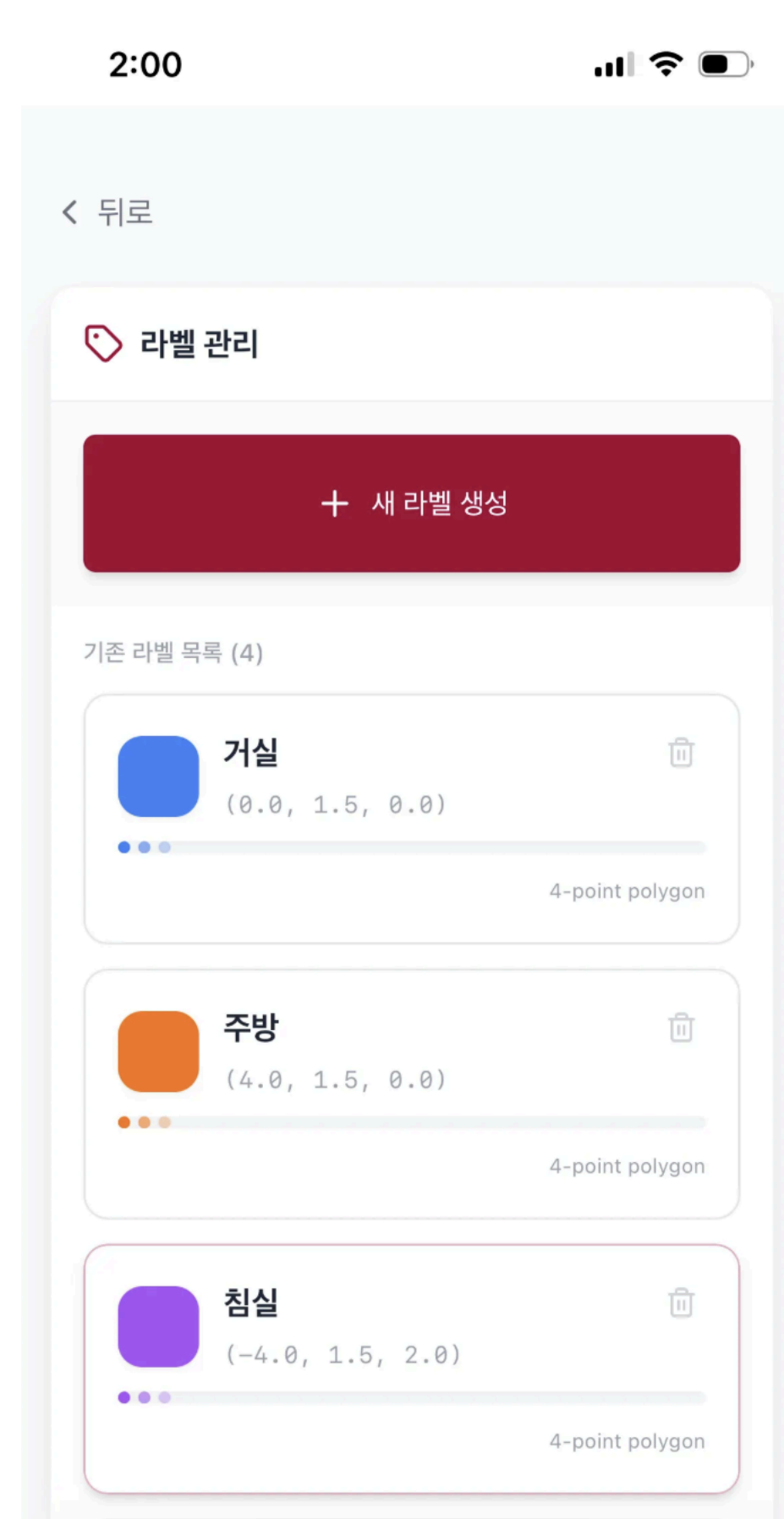
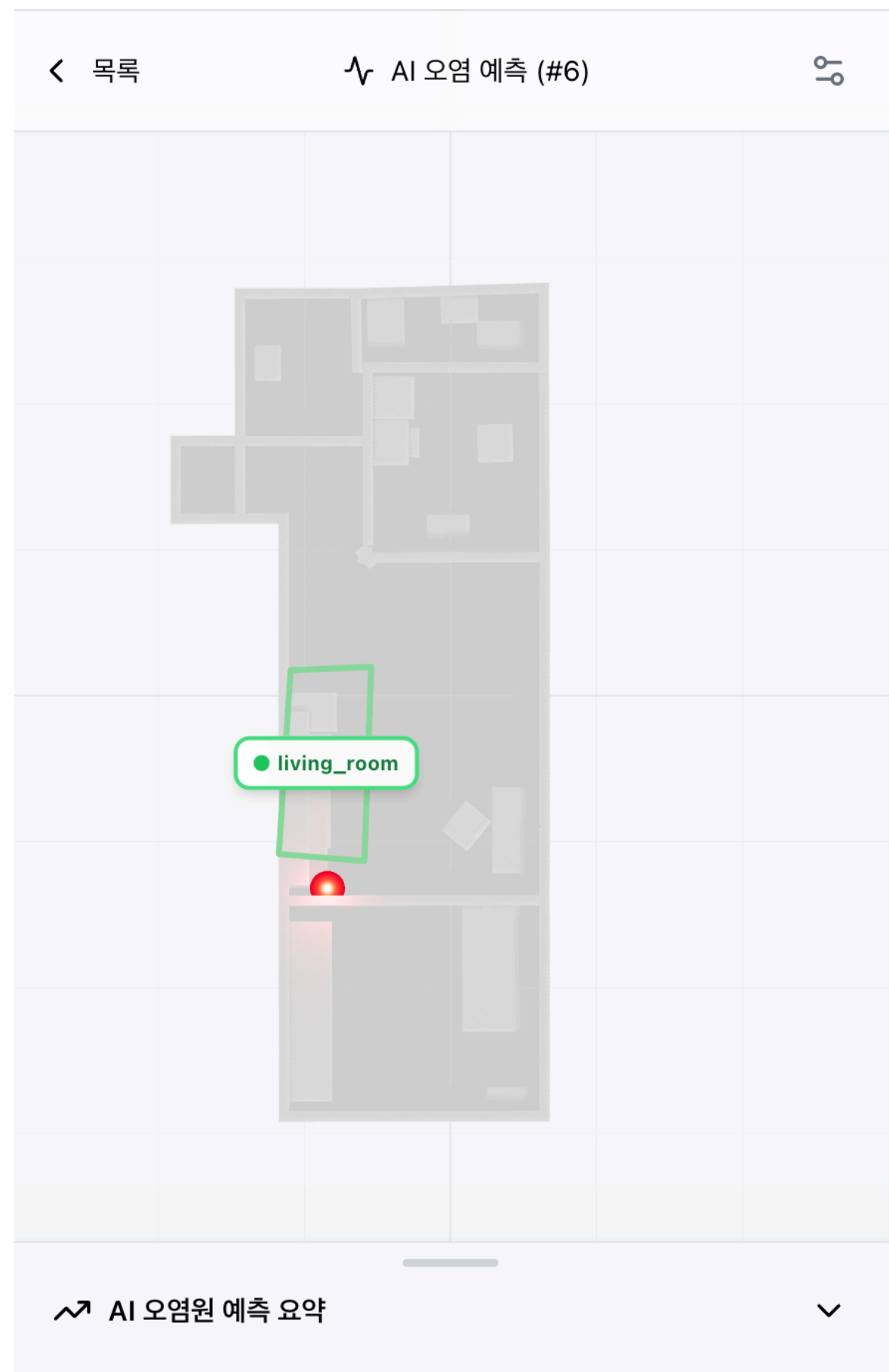
Architecture



Video

FR1

Home Structure & Location Intelligence Module



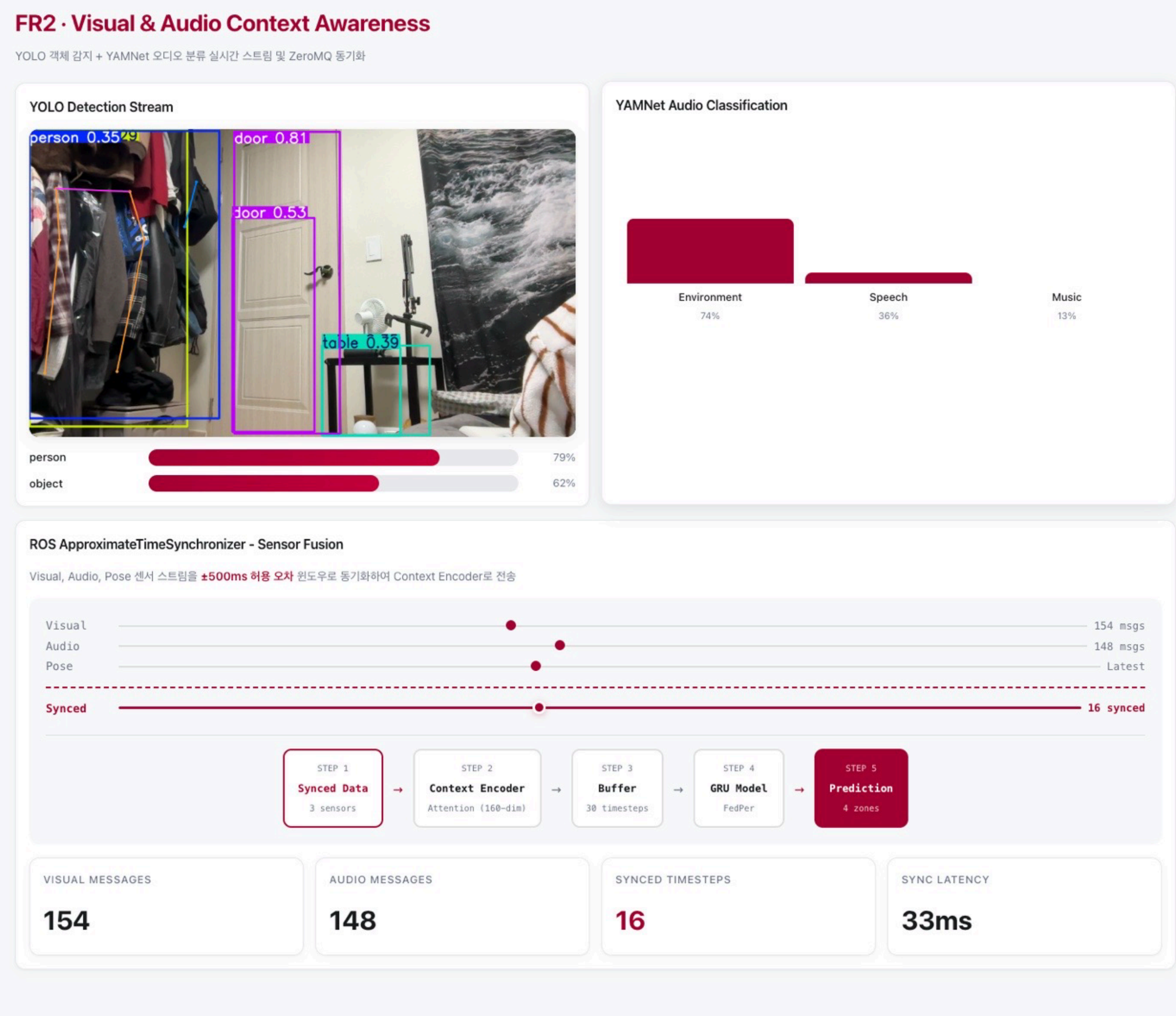
FR1

Home Structure & Location Intelligence Module

RobotLocation × + ⚙️							
🔄 Filters None Fields All Showing 8049 of 8049 Add record							
ℹ️ Use filters to narrow your search results. Multiple filters show results at their intersection (AND). + Add a new filter							
☐		deviceId #	robotMapId #?	recordedAt 📅	x #	y #	z #
☐		1	null	2025-12-01T20:15:47.7...	0.01202812325209379	0.02966070920228958	-0.05496197938919067
☐		1	null	2025-12-01T20:15:47.9...	0.07794220745563507	0.1142451092600822	-0.1421101689338684
☐		1	null	2025-12-01T20:15:48.1...	0.126288577914238	0.1716416776180267	-0.2275986075401306
☐		1	null	2025-12-01T20:15:48.3...	0.1492372900247574	0.2158086895942688	-0.2977194488048553
☐		1	null	2025-12-01T20:15:48.5...	0.1612328588962555	0.2361870259046555	-0.3698869347572327
☐		1	null	2025-12-01T20:15:48.7...	0.1624085158109665	0.2497199773788452	-0.4603172242641449
☐		1	null	2025-12-01T20:15:49.0...	0.1541064083576202	0.2456057369709015	-0.539044201374054
☐		1	null	2025-12-01T20:15:49.2...	0.1410153210163116	0.2298321127891541	-0.6009550094604492
☐		1	null	2025-12-01T20:15:49.4...	0.1264450550079346	0.2140495628118515	-0.6404898762702942
☐		1	null	2025-12-01T20:15:49.6...	0.1175685301423073	0.2118505239486694	-0.6592503190040588
☐		1	null	2025-12-01T20:15:49.8...	0.1102940589189529	0.2213513851165771	-0.6756537556648254
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☐		1	null	2025-12-01T20:15:50.6...	0.09052425622940063	0.226313903927803	-0.6977425813674927
☐		1	null	2025-12-01T20:15:50.8...	0.09857244044542313	0.2398770600557327	-0.6759548187255859

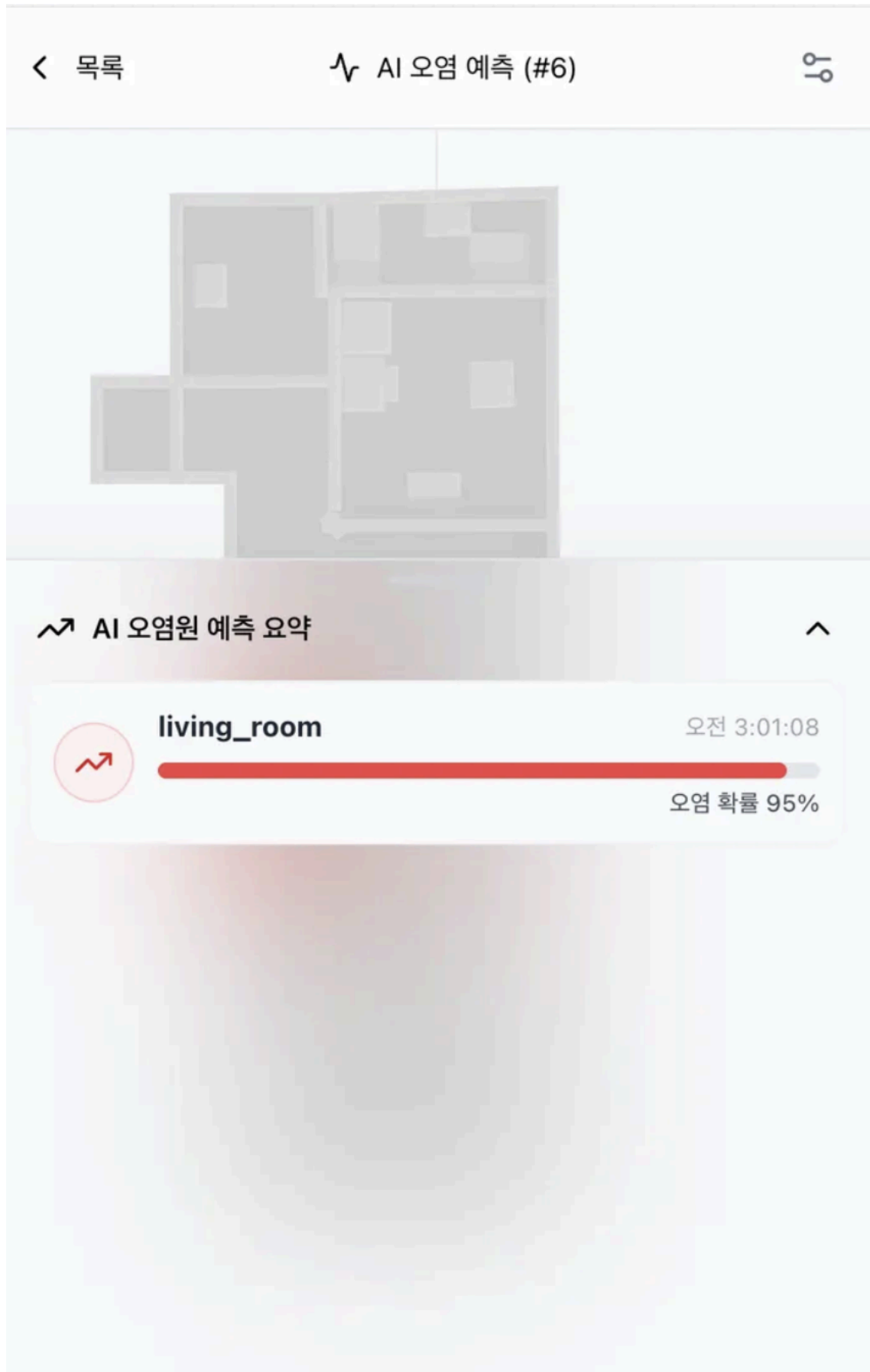
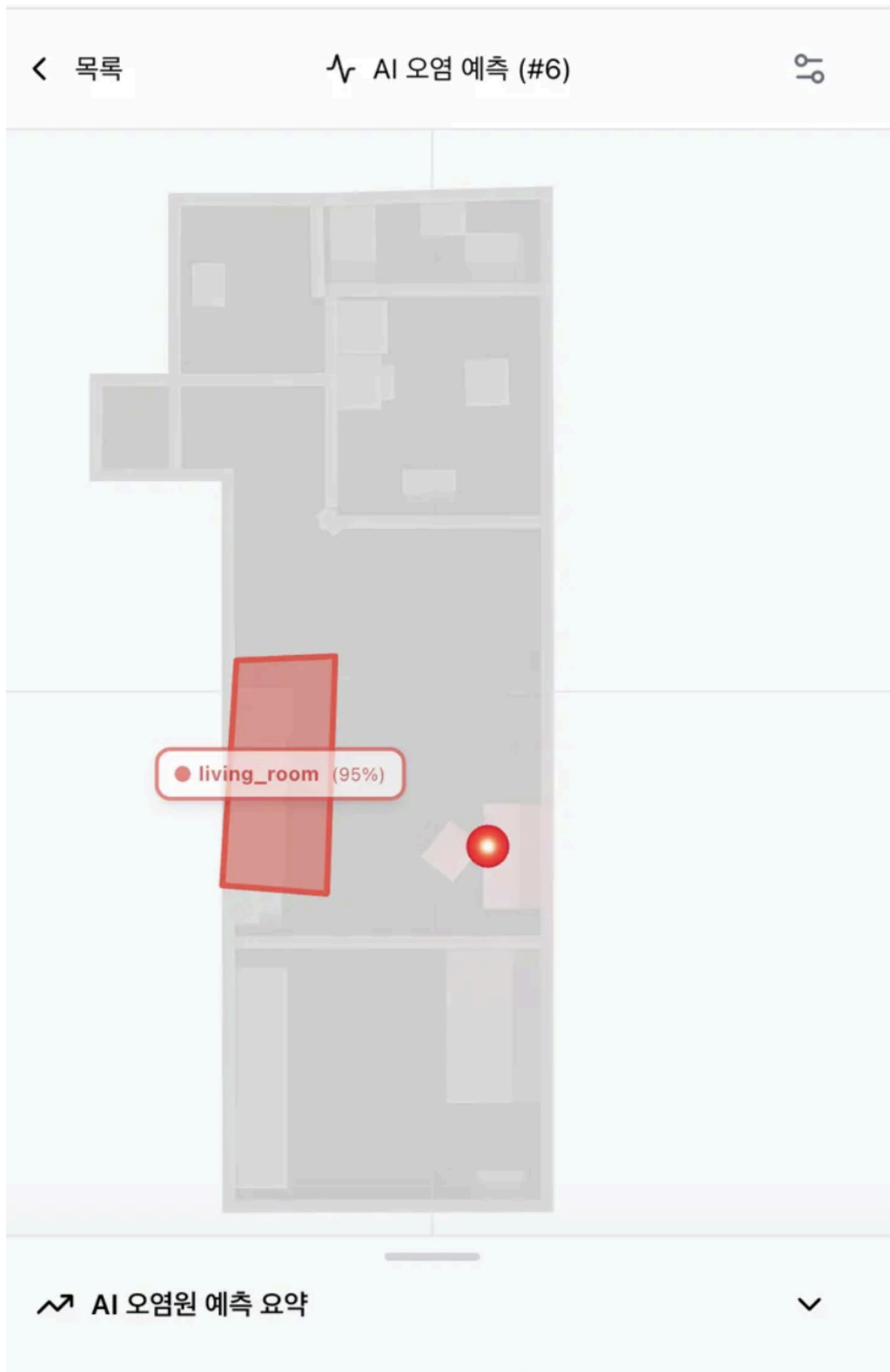
FR2

Visual & Audio Context Awareness



FR3

Sequential GRU prediction



FR3

Sequential GRU prediction

FR3 · Sequential GRU Prediction

30×160 Context Sequence → Attention Mechanism → GRU → Pollution Prediction

WebSocket Bridge Status

Connected to ZeroMQ Bridge

30×160 Context Sequence (Recent 30 minutes)

최근 30분 동안 집안 상황의 주요 변화 (30 timesteps × 160 features)

오전
5:58:20

spatial sensor
Processing...

오전
5:58:20

time sensor
Processing...

오전
5:58:20

time sensor
Processing...

오전
5:58:20

visual sensor
Processing...

오전
5:58:20

visual sensor
Processing...

오전
5:58:20

pose sensor
Processing...

오전
5:58:20

pose sensor
Processing...

Attention Weights

GRU가 예측에 가장 큰 영향을 준 특징들

Person movement pattern

92%

Object dropping sound

87%

Cooking activity

73%

Time of day

58%

Door events

45%

핵심 인사이트: 사람 이동 패턴이 예측에 가장 큰 영향을 주었습니다.

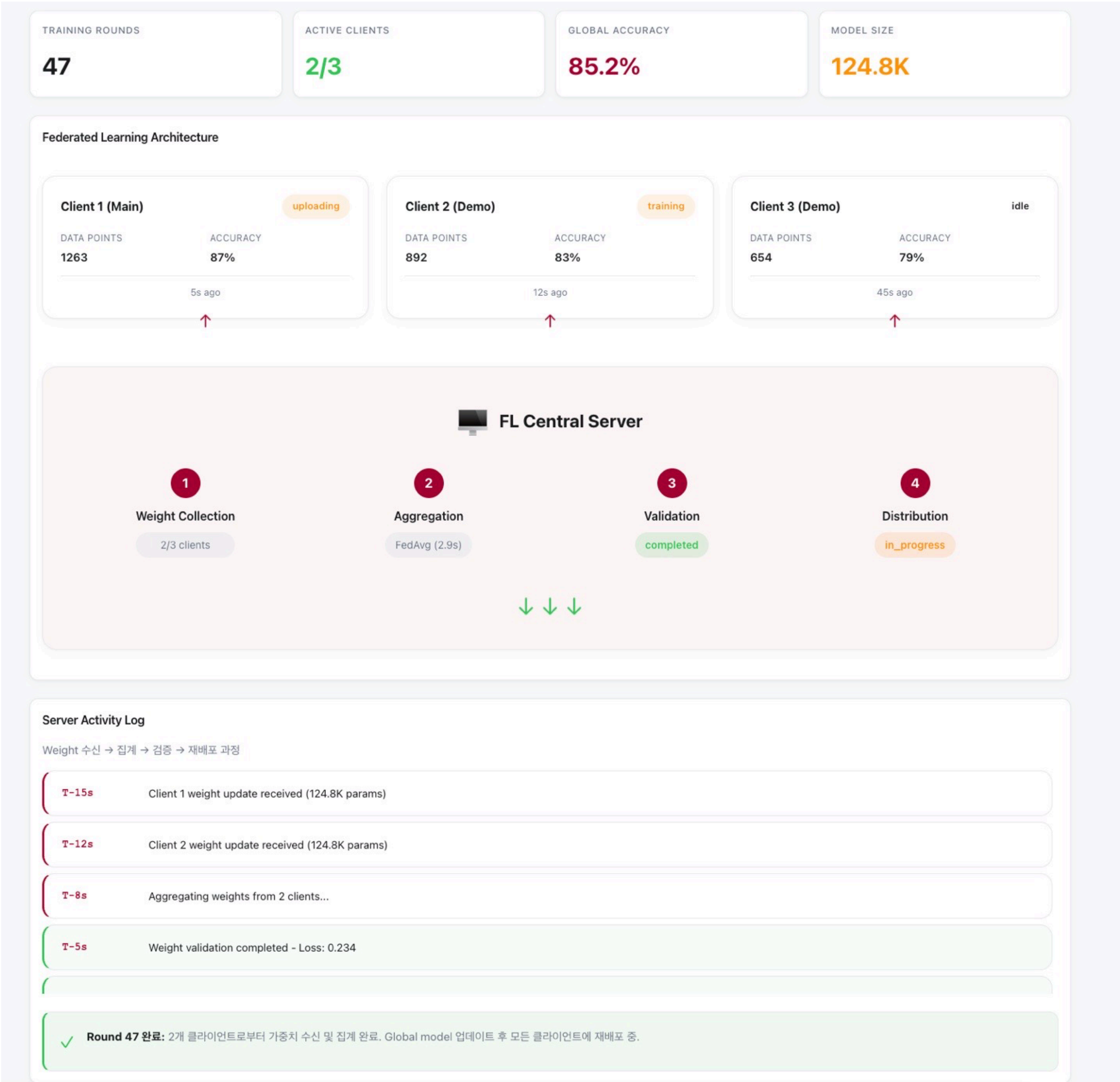
Pollution Prediction Results

구역별 오염도 예측 (0.0 ~ 1.0)

실시간 센서 데이터 수신 대기...

FR4

Personalized Federated Learning



FR5

Policy Engine & Cleaning Decision

FR5 · Policy Engine & Cleaning Execution

상황 인식 기반 정책 결정 및 청소 로봇 제어



CURRENT ACTION
청소 명령 대기 중...



13%

Policy Decision Reasoning

 거실에서 물건 떨어지는 소리가 감지되었고, 현재 조용한 시간대가 아니므로 즉시 청소를 시작합니다. 사용자 활동 패턴 분석 결과 현재 시간대가 청소에 가장 적합합니다.

QUIET HOURS

Inactive

USER PRESENT

Yes

BATTERY LEVEL

87%

LAST CLEANING

3 hours ago

Floor Map - Pollution Heatmap

구역별 오염도 시각화 (모바일 앱의 RoomPlan 기반)

Living Room

65%

Priority: #1

Kitchen

40%

Priority: #2

Bedroom

25%

Priority: #3

Hallway

15%

Priority: #4

High (>60%)

Medium (30-60%)

Low (<30%)

Cleaning Path Timeline

우선순위 기반 청소 경로 및 예상 시간

14 : 30

Living Room

Deep clean

15 min

Q

14 : 45

Living Room

Spot clean (detected spill)

5 min

O

14 : 50

Kitchen

Standard clean

12 min

O

15 : 02

Bedroom

Light clean

8 min

O

TOTAL TIME

40 min

REMAINING

35 min

CURRENT TASK

5 min

FR5

Policy Engine & Cleaning Decision

